

Umm Al-Qura University

Information Architecture Analysis for a Grocery Delivery

Application: A Case Study of the "Ninja" App

Course Code: Information Architecture (IA)

HCI2305

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1. Introduction

This report aims to analyze the user experience of the "Ninja" delivery application. An online survey was conducted to gather feedback from users of various age groups, particularly university students aged between 18-25 years. This analysis seeks to enhance the application's design and functionality based on actual user needs

1.1 Objectives

- Understand user behavior and patterns when using the Ninja app.
- Evaluate user satisfaction with the interface and services.
- Identify the main challenges or problems users face.
- Provide recommendations to improve the user experience and increase customer retention.

2. Phase One: Investigation (Data Collection)

2.1 Research Tool: Survey

[Click here to open the survey](#)

- **2.1.1 Methodology:** An online survey was created and distributed among a sample of users, most of whom were university students aged between 18-25 years. The survey included sections on general information, usage behavior, satisfaction with features such as navigation, product availability, pricing, and overall service quality.
- **2.1.2 Survey Results:** The results will be analyzed to identify the strengths and weaknesses of the application and provide recommendations for improvement.

2.2 Analysis

1- Would you like to continue this survey?

- 97% of users answered "Yes".
- A positive response reflects users' interest in improving the service, indicating their willingness to provide valuable feedback.

2 - User demographics (how old are you)?

Age: Most users (80%) are between the ages of 18 and 25. This is an important age group to understand well. Other age groups represent much smaller percentages.

3 - What is the nature of your current work?

- 84% of users are students.
 - 11% employees.
- The demographics indicate that the app primarily targets young people. Marketing campaigns should focus on the needs of this segment.

4 - Frequency of application use (approximately how often do you use the Ninja application?)

- 16% of participants are frequent app users (more than twice a week), indicating that a large percentage of users rely on the app regularly.
- 40.7% are average users (2-3 times per month), indicating that there is a large group of people who use the app moderately.
- 36.3% are infrequent users (once a month or less), indicating that there are users who use the app infrequently.
- 6.6% of participants are never users and have never used the app, reflecting a small segment of inattentive users.

5 - Types of requests?

The most popular orders are groceries (60%), followed by restaurants (30%). This indicates that the app is successfully meeting users' needs in these categories. Groceries stand out as a key category, indicating that users tend to use them for their daily needs, reflecting their reliance on the app to save time and effort. Although restaurant orders represent 31%, there is a significant opportunity to increase this share by improving offers and promoting available options.

6 - Why do you use the application frequently?

- 66% chose “convenience and speed of delivery.”
- 30% indicated “variety of options.”
- 25% indicated “good prices.”
- 33% indicated “easy to use.”

Convenience, speed of delivery, and ease of use are key factors, reflecting the importance of user experience when choosing an app.

7 - Is the product classification (such as vegetables and meat) clear and easy to navigate?

- 54.8% of users said "always".
- 40.3% said "often".

This indicates smooth navigation and easy finding of products.

8 - Did you encounter difficulty adding products to the cart or tracking your order?

- 64.3% said "No".
- 19% said "rarely".

Most users don't experience any difficulties, but any existing issues should be addressed to ensure a smooth experience.

9 - What interests you most?

Ease of ordering: 63% of users consider it the most important, indicating a need to improve the user experience.

Delivery speed: 50% also consider it important, indicating the importance of improving delivery efficiency.

Offers and discounts: 35% are interested in them, indicating that attractive offers can boost app usage.

Store variety: 30% consider it important, indicating users' desire for a variety of options.

10 - Is navigation between the application sections (page, home, categories, cart) clear and easy?

- Yes, always: 83.5%
- Sometimes: 15.3%

The vast majority (83.5%) of users find navigation between app sections clear and easy, indicating an effective user interface design. The 15.3% who indicate “sometimes” reflect the possibility of some issues that need to be addressed to improve the user experience.

11 - Did you have difficulty understanding the category names?

- 55% indicated that they did not face any difficulty.
- 16.3% said "rarely"
- 20% "sometimes"
- The majority do not face difficulties in this matter.

12 - How do you rate the ease of finding products?

- 40% said "easy".
- 33.3% said "always".
- 16% said "often."

It indicates that finding products is clear, easy and fast.

13 - What is the most prominent problem you faced while using delivery applications?

- 47% of users stated that “product unavailability” was the main problem.
- 21% reported “delay in delivery.”
- 12% reported “problems with payment.”

Focus should be placed on resolving product availability issues, improving delivery speed, and simplifying the payment process to ensure a positive user experience.

14 - How would you describe the payment steps in the app?

Possible outcomes:

The vast majority indicated the ease of the payment process, which reflects users' satisfaction with the application.

- If the largest percentage indicates "easy," this means that the checkout experience is good, but you should focus on improving the aspects rated as "medium" or "difficult" to ensure greater ease for all users.

15 - Have you ever left the app without completing an order? Why?

Possible outcomes:

1. Yes:

- Payment difficulties: If there are problems during the payment process, this may lead to users leaving.
- Product unavailability: If users don't find what they're looking for, they're likely to leave the app.
- Delayed delivery: If users feel that the delivery process will be slow, they may decide not to complete the order.
- User interface: If the payment or browsing steps are complex, it may lead to users being frustrated and exiting the application.

2. No:

- Conclusion: Users who did not exit indicate their satisfaction with the overall experience, which reflects the quality of service and effectiveness of the app.

2.3 Evaluation

Strengths in Design and User Experience

1. Clear Brand Identity

- The site uses consistent colors and a simple, recognizable logo that strengthens its branding.

2. Easy Navigation

- The main menu is clear and allows users to easily access different categories like “Cats” and “Dogs.”

3. Smooth User Experience

- Browsing products and adding them to the cart is simple. Multiple payment methods are supported, including Visa, MasterCard, and Apple Pay.
4. Available Customer Support
- Multiple communication channels with customer service help build trust with users.

Weaknesses and Areas for Improvement

1. Lack of Visual Clarity in Some Elements
- Some text is small or lacks sufficient contrast with the background, making it harder to read on certain devices.
2. Limited Product Information
- Some products lack detailed descriptions, which can affect the user's confidence in making a purchase.
3. No User Reviews or Ratings
- The absence of a review or rating system reduces the credibility of products and makes it harder for customers to make informed decisions.
4. Mobile Optimization Needs Work
- Although a mobile app exists, the website experience on smartphones could be improved in terms of loading speed and responsive design.

2.4 User Personas

1. Sara Al-Juhani - 43 years old

Gender: Female

Marital Status: Married

Occupation: Housewife Location: mecca

Needs from Ninja App:

- Fast and reliable delivery of daily household items
- Cash on delivery or Mada card payment
- Quick Arabic customer support
- Real-time delivery notifications

Story:

Sara is a mother of three who spends most of her day managing the household. She relies on Ninja because she can't leave the house often. She appreciates the real-time updates and consistent delivery times.

Noura Al-Ghamdi - 21 years old

Gender: Female

Marital Status: Single

Occupation: University student

Location: Near King Abdulaziz University - Jeddah

Needs from Ninja App:

- Easy and fast to use
- Discounts on snacks and light meals
- Scheduled deliveries during classes
- Delivery to campus buildings

Story:

Noura is a third-year student who likes to simplify her hectic schedule. She uses Ninja to order snacks or coffee during long lecture days. The app helps her a lot with frequent offers and convenient campus delivery.

Khaled Al-Mutairi - 29 years old

Gender: Male

Marital Status: Single

Occupation: Customer service employee at a bank

Location: Mecca

Needs from Ninja App:

- Accurate order tracking
- Neat and high-quality delivery
- Flexible online payment options
- Service availability during and after work hours

Story:

Khaled works a fixed 8–5 job and rarely has time to shop. He relies on Ninja for office snacks and small essentials. He values timely, presentable deliveries, especially when ordering for work.

3.1 Card Sorting

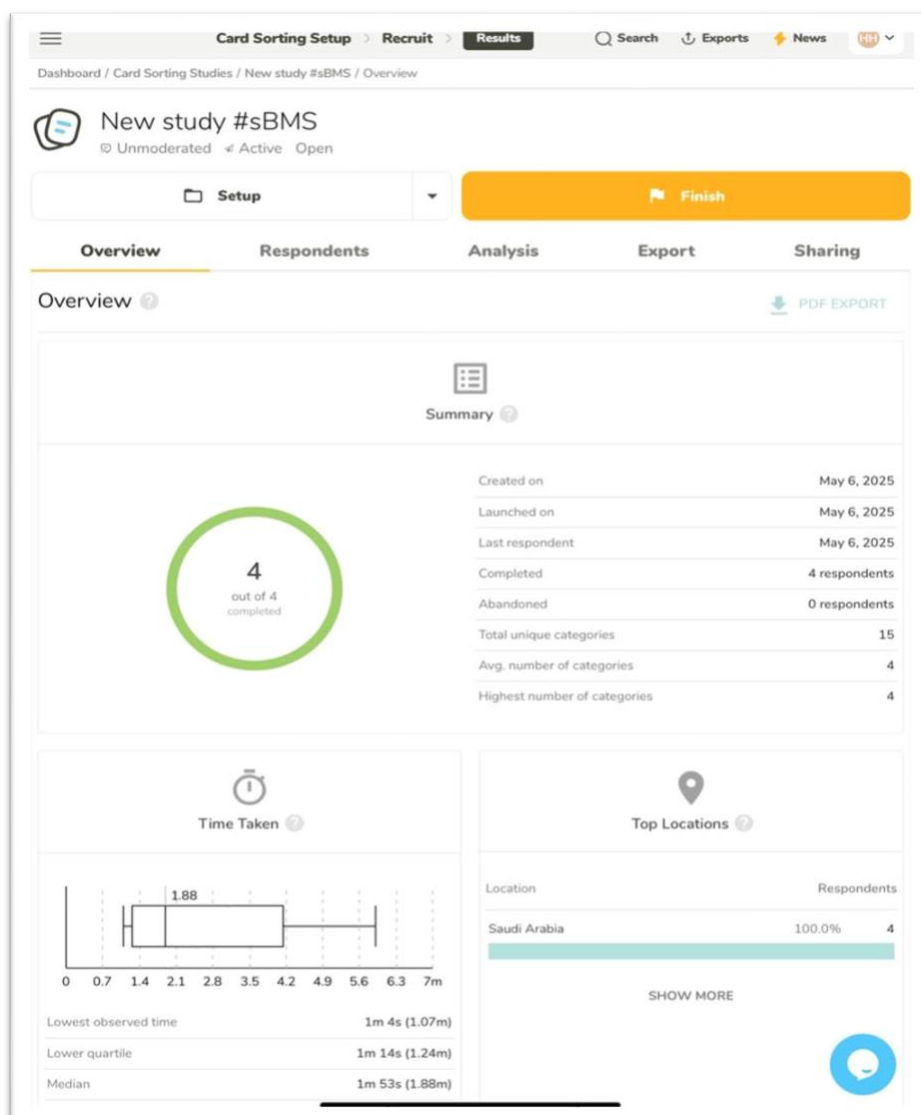
Card Sorting Activity

Participants were invited to take part in an open card sorting task to help structure the app's information architecture.

3.1.1 card sorting activity:

<https://study.uxtweak.com/cardsort/7eholA5VRY0U1IUUOX5IS>

3.1.2 Card Sorting Results:



3.1.3 Card sorting analysis

1. Onion

- Category: Vegetables
- Freq: 1
- Avg. Rating: 3.0
- Analysis:

Onions were rated only once, indicating low user engagement or limited choice. The average rating (3.0) indicates that it's a well-known ingredient but not a strong favorite, and may indicate that it's a staple but not always a top choice.

2. Cucumber

- Category: Vegetables
- Freq: 1
- Avg. Ratings: 4.0, 2.0, 5.0 → Approximate Average: 3.67
- Analysis:

The wide variation in ratings indicates a difference in participants' views on the option's importance. Its overall rating is good but not very high, indicating that it is accepted as a familiar component but not essential for everyone.

3. Garlic

- Category: Vegetables
- Freq: 1
- Avg. Ratings: 1.0, 3.0, 4.0, 5.0 → Approximate Average: 3.25
- Analysis:

Garlic is a well-known and essential ingredient in cooking. Although ratings varied, a low rating (1.0) indicated that some participants considered it a very important ingredient. The variability suggests that garlic use may depend on cultural background or personal preference.

4. Carrot

- Category: Vegetables
- Freq: 1
- Avg. Rating: 1.0
- Analysis:

Carrots' rating of 1.0 indicates that they are considered a staple and highly favored ingredient among participants. This demonstrates strong agreement on their nutritional importance and widespread use.

5. Broccoli

- Category: Vegetables
- Freq: 1
- Avg. Ratings: 4.0, 2.0, 1.0 → Approximate Average: 2.33
- Analysis:

Broccoli is often viewed as a healthy ingredient, but its varying ratings indicate that some participants dislike it despite recognizing its value. It is considered a good ingredient, but not a top favorite.

6. Mango

- Category: Fruits
- Freq: 1
- Avg. Ratings: 2.0, 1.0 → Approximate Average: 1.5
- Analysis:

Mangoes received a high rating, indicating their popularity as a favorite fruit. The classification as a fruit was clear, reflecting a collective perception that it is a beloved staple fruit.

7. Beef

- Category: Meat and Protein
- Freq: 1
- Avg. Rating: 1.0
- Analysis:

A low rating (1.0) indicates that meat is viewed as a primary and essential source of protein. The absence of alternative ratings reflects the

8. Watermelon

- Category: Fruits
- Freq: 1
- Avg. Rating: 2.0
- Analysis:

Considered a refreshing summer fruit. Fairly high rating but not a core staple like mango.

9. Milk

- Category: Dairy Products
- Freq: 1
- Avg. Rating: 3.0
- Analysis:

Important dietary item, though not highly rated. Classification is accurate, but low frequency weakens the trend.

Problems Identified

1. Low Frequency

Each item was rated only once (Freq: 1), reducing statistical reliability.

2. Variation in Evaluations

Items like Cucumber and Garlic had large rating discrepancies, showing a lack of consensus.

3. Ambiguity in Classifications

Unclear codes like “W,” “H,” or “G” appeared without explanation, causing confusion.

4. Poor Representation

Categories like Dairy and Meat had only one item, limiting insight.

5. Inconsistent Analysis

Some item analyses lacked depth or differentiation, e.g., Carrot vs. Garlic.

Recommended Improvements

1. Increase Participants

Larger samples will provide more reliable, generalizable results.

2. Reorganize Data

Use structured tables to display ratings/frequency for comparison.

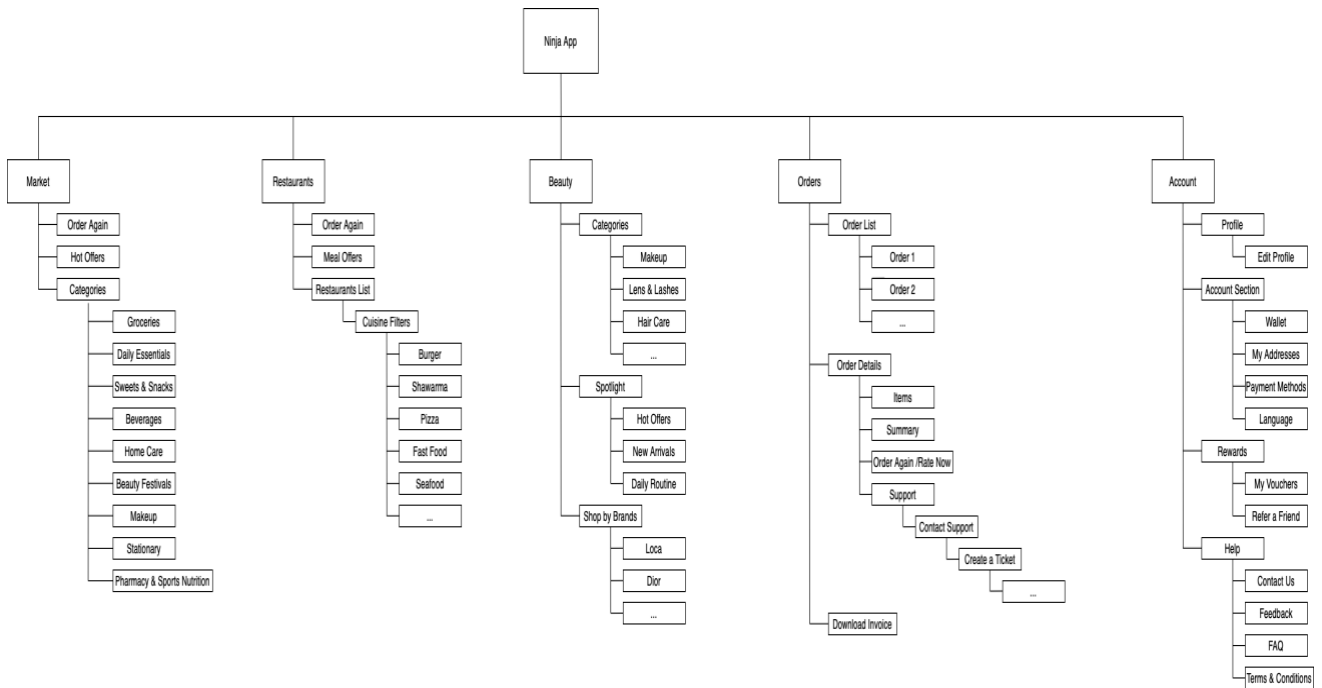
3. Deeper Analysis

Use descriptive statistics (mean, standard deviation), and compare items within categories.

4. Test Categories Separately

Run individual card sorts for each category (vegetables, fruits, etc.) to understand sorting logic.

3.2 Sitemap Creation



3.3 Navigation & Search Design

1. Navigation Design

Based on findings from Phase 1, users highly value ease of navigation and quick access to key features. Therefore, we propose the following:

- Implementing a Bottom Navigation Bar with key sections:
 - Home
 - Store
 - Offers
 - Recipes
 - My Account
- Providing a simple and intuitive navigation system that aligns with users' mental models.
- Adding a secondary (hamburger) menu for additional settings or subcategories if needed.

2. Search Experience Enhancement

Some participants noted occasional difficulties in locating products. To improve this experience, we suggest:

- **Autocomplete** suggestions during typing.
- **Synonym recognition** for example: ("tomato" and "tomatoes").
- **Search filters** based on:
 - Price range
 - Product type for example: (organic vs regular)
 - Availability (in-stock items only)

Connection to Research Findings

- 63% of users prioritized ease of ordering, highlighting the importance of seamless navigation and effective search.
- 54% confirmed that product classification was clear, but enhancements could still improve efficiency.
- Product unavailability was reported as the top issue (47%), suggesting improved search and filtering may help reduce user frustration.

3.4 Wire framing

